



DEVELOPMENT OF AN INTELLIGENT RECOMMENDATION SYSTEM IN TOURISM USING NATURAL LANGUAGE PROCESSING AND HYBRID ALGORITHMS

Mehtap KORKMAZ

Supervisor: Prof. Dr. Ecir Uğur KÜÇÜKSİLLE

INTRODUCTION

A hybrid decision support and recommendation system architecture was developed to analyze user experiences in Antalya hotels through semantic and statistical dimensions. Large-scale user feedback was processed using a layered engineering framework that combines statistical methods with modern LLMs. Traditional numerical ratings were challenged by extracting deep semantic sentiment intensities from unstructured text. Consequently, traditional data analysis workflows were transformed into an interactive and stable software product for end-users.

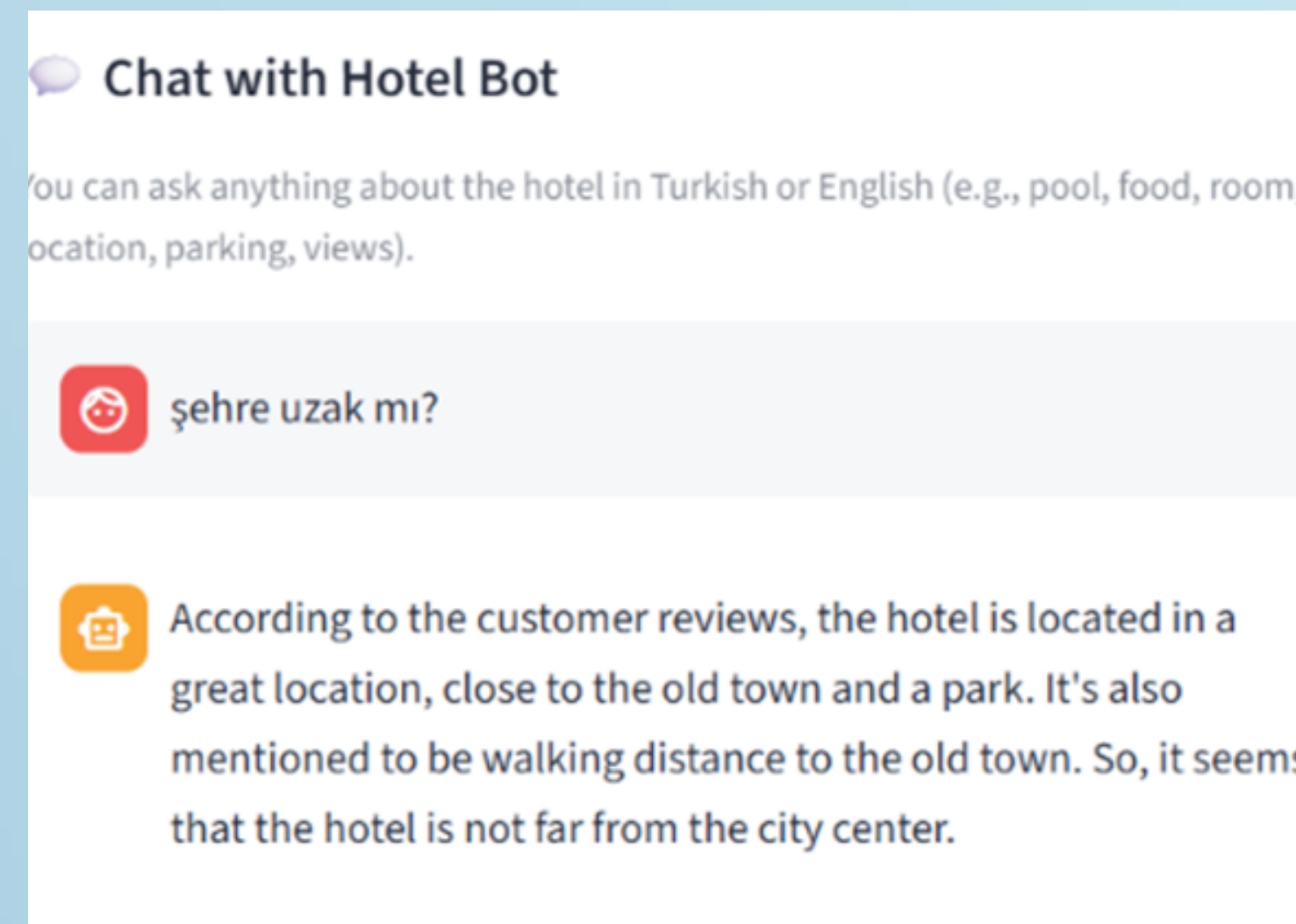
MOTIVATION

- Addressing Information Asymmetry: Resolves the conflict between high numerical scores and negative textual feedback, ensuring travelers make informed decisions.
- Semantic Intelligence (Llama 3.1 & VADER): Integrates VADER for precise sentiment scores and Llama 3.1 to process thousands of past guest comments to understand complex, natural language queries.
- Objectivity: The system filters out biased ratings by recalculating a hotel's rank based on actual sentiment intensity derived from raw text.
- Usage Areas: Tourists seeking verified recommendations, travel agencies requiring data-driven insights, and hotel managers tracking guest sentiment trends.

MATERIAL & METHODS

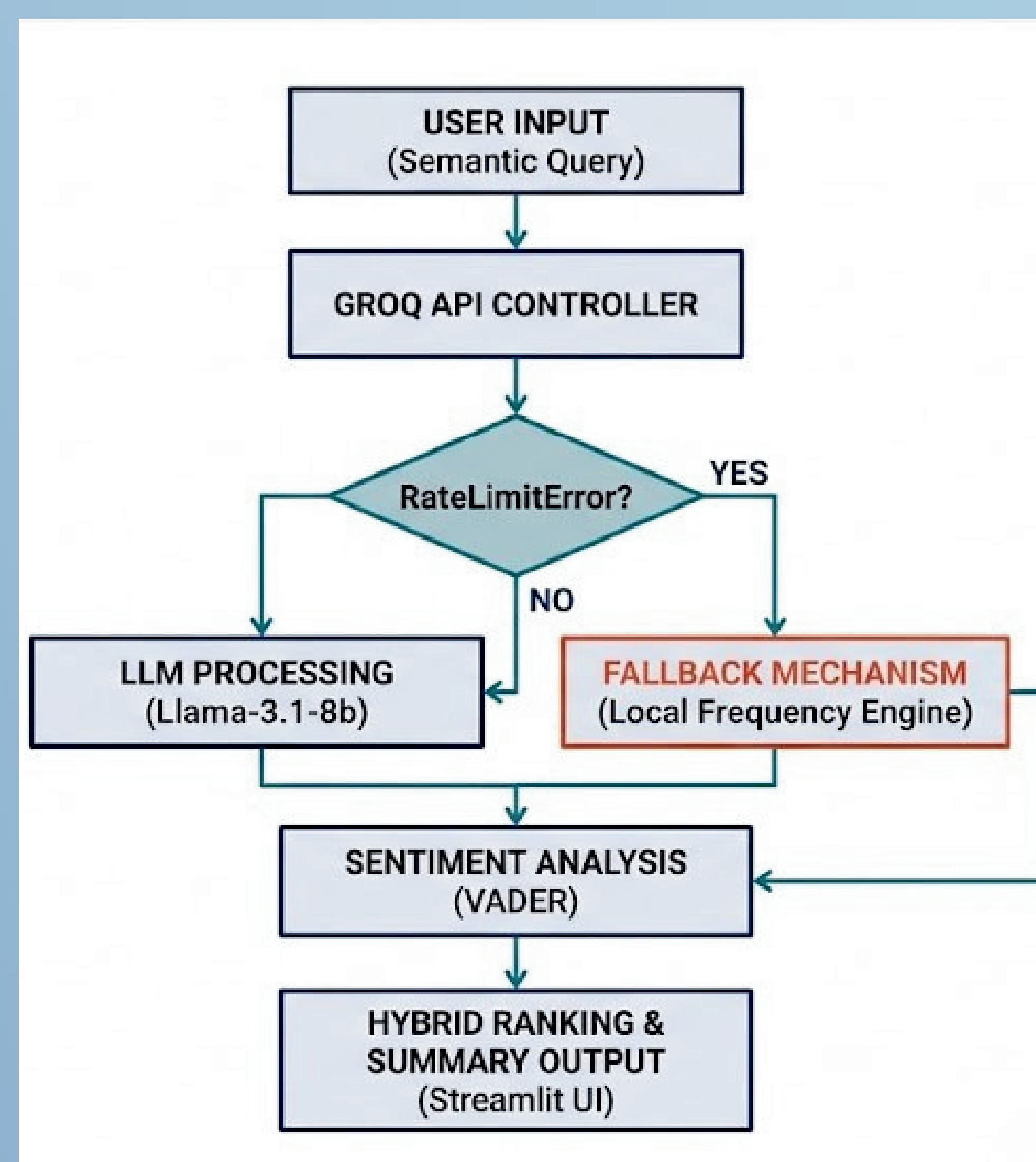
- Data Collection & Pre-processing: A curated dataset consisting of 4,000+ user reviews from 195 Antalya hotels was manually filtered for noise. Python-based cleaning pipelines ensured data integrity.
- The VADER: Valence Aware Dictionary and Sentiment Reasoner algorithm was integrated to calculate compound sentiment scores for each review, providing the qualitative weight for the hybrid model.
- Generative AI Integration: The Llama-3.1-8b-instant model was deployed via the Groq API to perform abstractive summarization and handle semantic natural language queries.
- Environment & Deployment: The entire application is containerized using Docker (Python 3.13.5-slim) and deployed via Streamlit on Hugging Face Spaces.

RESULTS



ARGUMENT

- Project Goal Achievement: An operational, hybrid decision support prototype has emerged.
- Unfinished Components: There are no missing core functionalities in the system, though the underlying hotel review dataset is currently static.
- Encountered Difficulties: The main challenge was cloud API rate limiting, which was resolved by engineering a custom fallback mechanism.
- Expected vs. Actual Results: The system performed exactly as expected, effectively recalculating hotel rankings by matching quantitative metrics with text sentiment.
- Future Work & Scalability: Future stages can integrate real-time automated data scraping pipelines and a Vector Database to implement a full Retrieval-Augmented Generation (RAG) framework.



THANKS

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